OPTIMIZING DATA STRUCTURES WITH ALTERNATING-ORIENTED BISECTED LAYOUTS

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1 OVERVIEW

Layouts are a general and universal design tool. However, for the purposes of modern computing layouts are lacking a general and universal data structure. In this paper, a layout data structure will be described and applied to various areas of computer science. It will be proposed that there is a data structure that can be thought of as a general theory of layouts.

The XOO-layout is a general-purpose and scalable tool for design logic. For the purposes of determining and optimizing data structures, the XOO-schema is constrained by bisecting divisions of the layout in alternating orientations.

There has been some interesting research in layout and applications of layout structure over the years and with a little bit of insight it extends into many areas of logic. One of the interesting properties of the XOO-schema in applying a loose and strict set of rules to its data structure is that interesting geometries appear and that there are overlaps in many areas of computer science.

16	18	24	26
17	19	25	27
20	22	28	30
21	23	29	31

Figure 1: 4-Dimensional Dense Layout

1.1 APPLICATIONS

There are a number of areas where this layout engine can be applied:

- Design logic

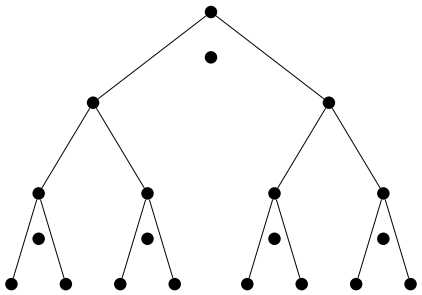


Figure 2: Binary Hypertree

- Circuit design
- Parallel computing
- Distributed computing
- Distributed networking
- Software development
- Manufacturing
- Architecture

1.2 IMPLEMENTATION

To begin, a general layout data structure is described. This is done in an iterative approach. This is as specified in "a method for users to create and edit web page layouts." [84]

Begin with a rectangular region R_1.

Bimedians H or V bisects the region R_1 into equal subregions R_2 and R_3. Only one bimedian is allowed per region and bimedians alternate between H and V in each subregion relative to its region.

In general,

$$R_i \rightarrow R_{2i} \cup R_{2i+1}$$

and

$$R_{2i} \cap R_{2i+1} = \emptyset$$

The region R_1 is also known as a topological flat torus.

The initial region is bisected horizontally or vertically.

Figure 4 is horizontally bisected. Figure 5 is vertically bisected. Figure 6 illustrates a general, open-ended approach to layouts. Each subdivision of a layout can only be bisected either horizontally or vertically.

Start with an empty layout, initially O, labeling the subdivisions by its binary tree encodings.

Graphing the layout is done in the following way. Subdivisions are bisected alternating between horizontal and vertical orientations. Each subdivision is exactly half of its parent in the orientation of the bisection. Each subdivision will be represented by a graph vertex. A graph edge will represent neighbourliness of subdivisions as represented by the layout displayed.

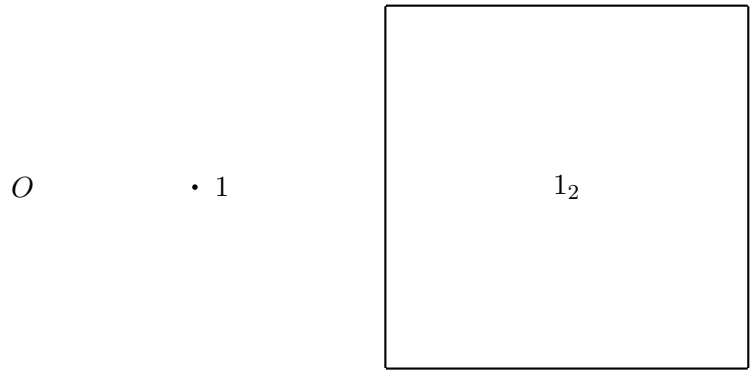


Figure 3: Initial Division

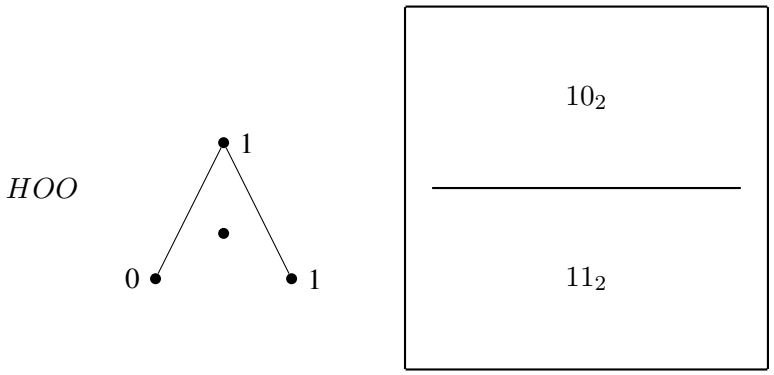


Figure 4: Horizontal Subdivision

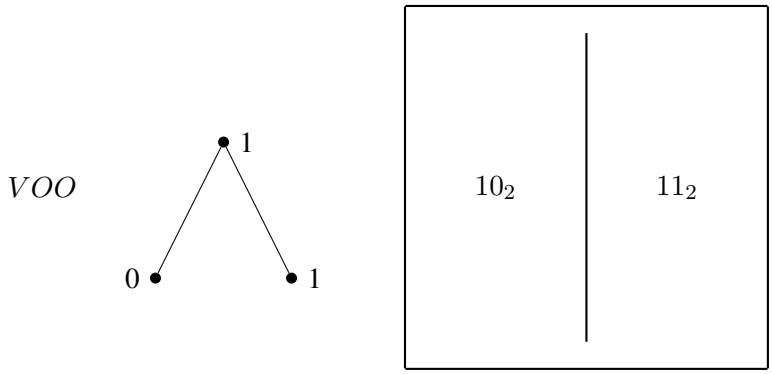


Figure 5: Vertical Subdivision

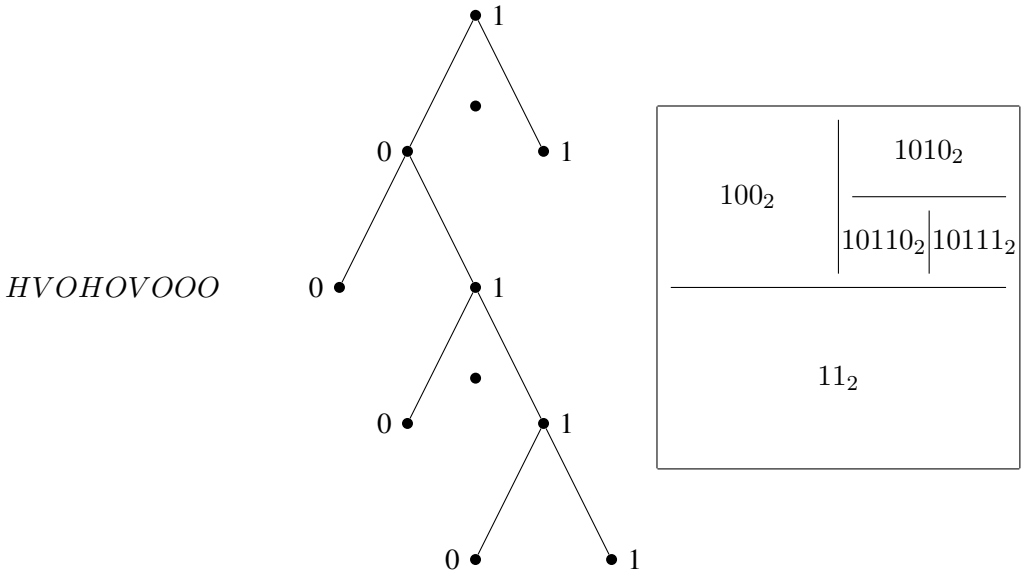


Figure 6: Multiple Subdivisions

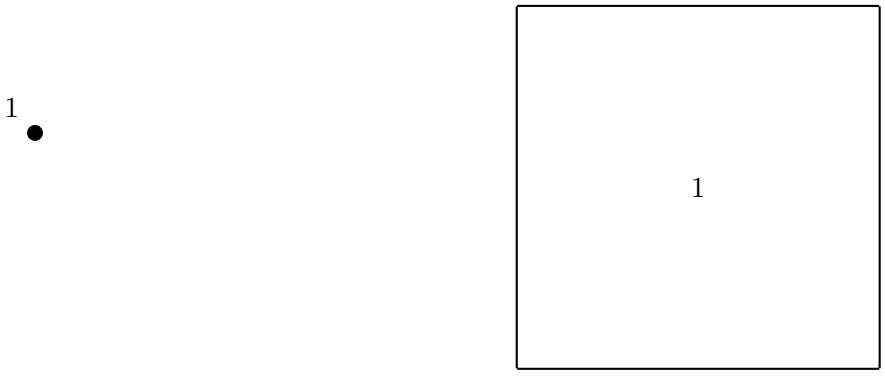


Figure 7: Initial Layout and Graph

The initial division has no neighbours so there is only one vertex and no edges. This is a zero dimensional XOO-encoding.

Next, subdivide the layout into VOO and label the subdivisions according to the XOO-layout. Be sure to subdivide each division only once and when this is done this is called a dense layout - no more divisions can be subdivided without increasing the dimension. Figure 8 is a 1-dimensional layout and graph.

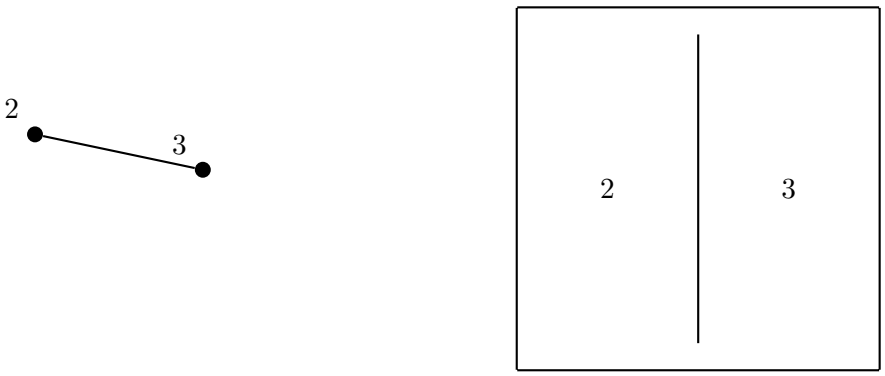


Figure 8: VOO Layout and Graph

Then subdivide the VOO layout this time replacing each O with HOO. This dense layout is then 2-dimensional.

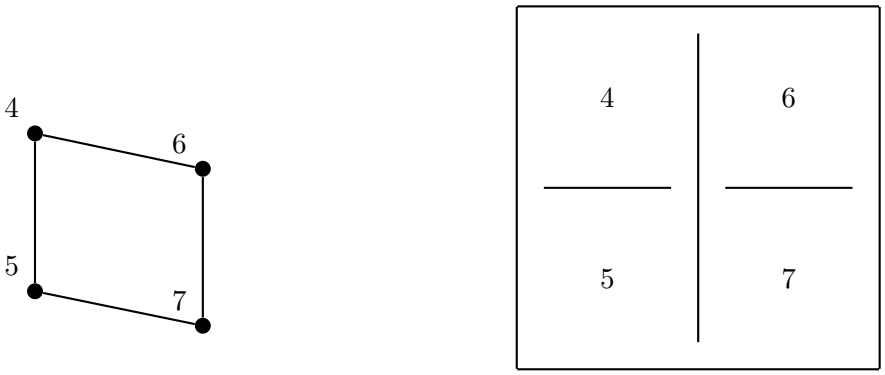


Figure 9: $VHOOHOO$ Layout and Graph

Continue on alternating between VOO and HOO subdivisions and the overall dense layout grows by one dimension.

As can be seen, graphing the layout in such a way as having subdivisions represented by vertices and neighbours representing edges, it begins to represent a 3D cube. This means that a two-dimensional XOO-layout can represent a n-dimensional graph.

Continuing with the next subdivision, a 4D layout and graph are represented.

This continued subdivision of dense layouts represents hypercubes.

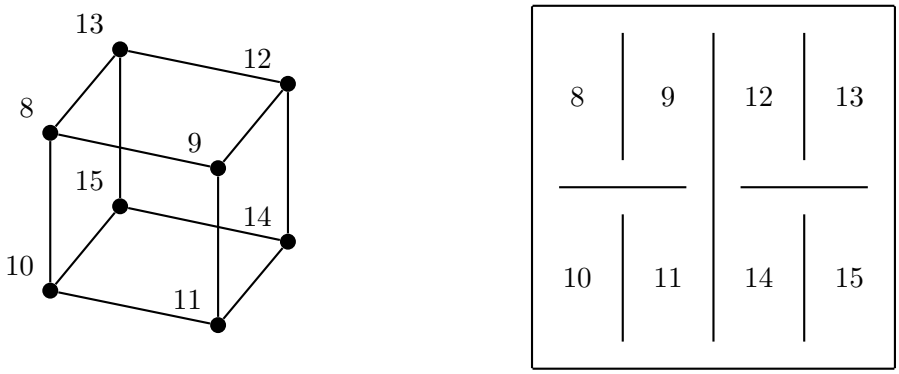


Figure 10: $VHVOOVOOHVOOVOO$ Layout and Graph

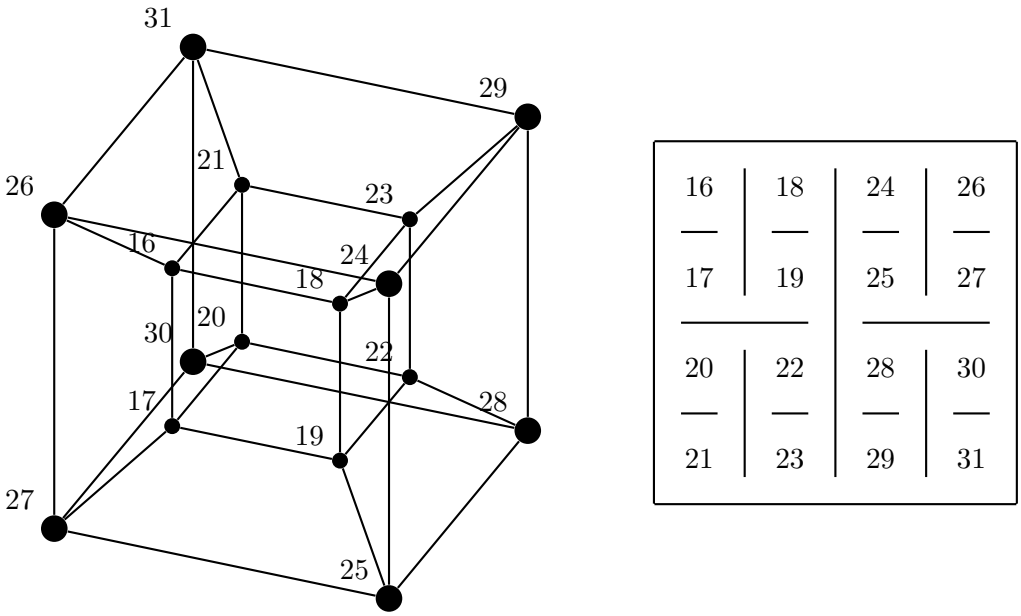


Figure 11: $VHVHOOHOOVHOOHOOHVHOOHOOVHOOHOO$ Layout and Graph

This extension of "a method for users to create and edit web page layouts" [84] includes an implementation regarding the encoding of the XOO-encoding. Essentially, the XOO-encoding is a specialized ternary encoding. The three characters (O, V and H) can be encoded in binary using Huffman encoding as a basis. So, the most frequent character O represents approximately half of all the XOO-encoding so it's encoded as 0_2 (zero). The character V and H appear equally frequently so will be represented as 10_2 and 11_2 respectively.

$$O \equiv 0_2 \equiv 0_3$$
$$V \equiv 11_2 \equiv +_3$$
$$H \equiv 10_2 \equiv -_3$$

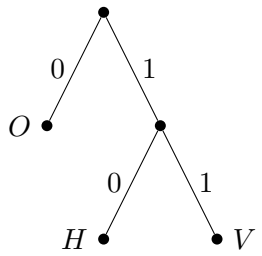


Figure 12: Symbol Encoding Using Huffman Encoding

These encodings are equivalent:

$$HVOHOVOOO$$
$$-+0-0+000$$
$$1\,0110\,1001\,1000_2 = 0x1698$$

A benefit of the XOO-encoding is compression as ternary encoding is more compact and efficient than binary encoding. The XOO-encoding can then be used to compute sophisticated data structures very efficiently.

Another benefit of this data structure is that data structures can be free of variable names. All of the structure and node information can be extracted from the encoding.

1.3 DESIGN LOGIC

"A method for users to create and edit web page layouts" describes how layouts are encoded, represented and displayed with a user interface for editing. It is extended as above using graphs to represent hypercube architectures and various applications as below.

1.4 CIRCUIT DESIGN

In circuit design, Karnaugh maps are an established way of simplifying truth tables representing circuits. [86]

Karnaugh maps purpose is to simplify real-world logic so that circuits can be implemented using a minimum number of physical logic gates.

Starting with a simple 4-input circuit (Figure 13), the truth table representing the circuit logic is shown (Figure 14).

The Karnaugh map, or K-map, is a tabular representation of the logic table. The K-map implements Gray codes as part of the tabular representation. In contrast, using the XOO-layout eliminates the need for explicit Gray codes and eliminates the need to look for groups of 1s wrapping around the K-map.

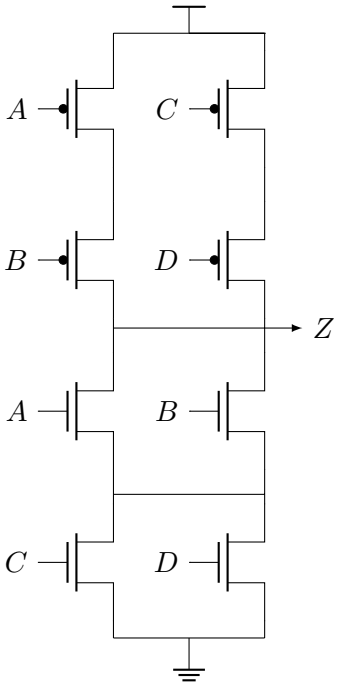


Figure 13: Circuit Diagram

A	B	C	D	Z
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	0

Figure 14: Logic Table of Circuit Diagram of Figure 13

Using the XOO-map requires only identification of groups of 1s and any symmetries horizontally and vertically (but not diagonally). The XOO-map can merge similarly adjacent 1s and 0s, and group those together in larger subdivisions. Then finally, reading off the XOO-map provides the simplified sum-of-products. Additionally, it can also be represented variable-free in the XOO-encoding.

The alternative XOO-map (Figure 15) is shown as defined by the truth table (Figure 14).

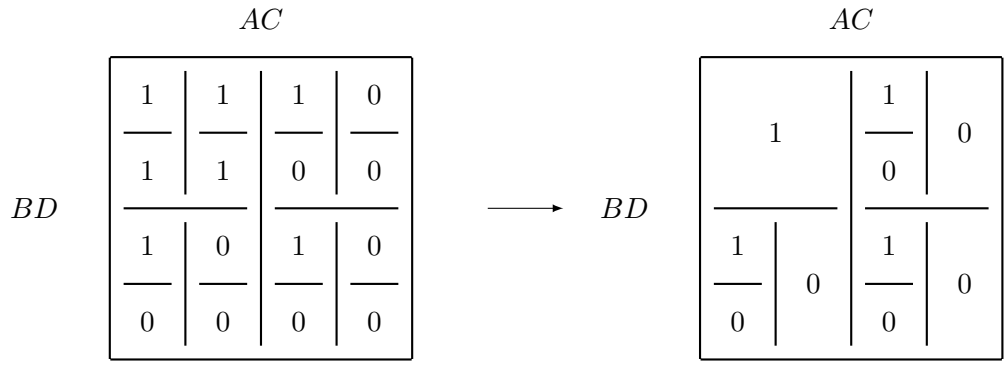


Figure 15: XOO-Layout Minimized

Reading the XOO-map takes some practice. Note that $\overline{A}$ equals $NOTA$, $\overline{B}$ equals $NOTB$, etc. Inputs alternate in a binary fashion.

By group, read the first upper-left quadrant group of 1s as $\overline{AB}$ because the group spans only $\overline{A}$ and $\overline{B}$. (It also spans C and $\overline{C}$, D and $\overline{D}$, so are ignored.) By symmetry, read the upper-left 1s of each quadrant as $\overline{CD}$ because the group spans only $\overline{C}$ and $\overline{D}$. (It also spans A and $\overline{A}$, B and $\overline{B}$, and so are ignored.)

Therefore, the sum-of-product is:

$$\overline{AB} + \overline{CD}.$$

The XOO-encoded, variable-free, sum-of-product is:

$$VHOVHOOOHVHOOOVHOOO, 1, 1, 0, 0, 1, 0, 0, 1, 0, 0$$

or (where O = 0_2, H = 10_2 and V = 11_2),

$$0x39C2E1C324.$$

The XOO-encoding is 40 bits. The original truth table is encoded in 5*16 = 80 bits (not including the labels.)

The graph of the above sum-of-products is a $sub - hypercube$ (Figure 16).

1.5 LOGIC GATES

Logic gates are easily interpreted in the XOO-layout producing some simplified logic tables.

The AND gate (Figure 17) is simplified to 10 bits:

$$VOHOO, 0, 0, 1.$$

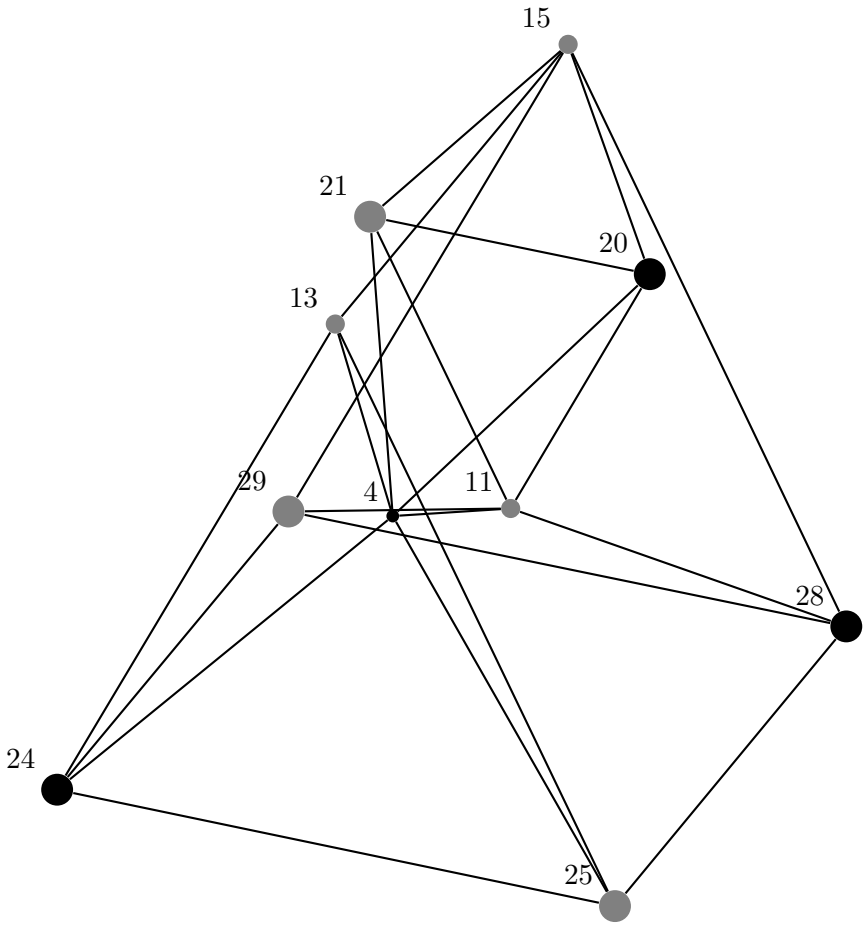


Figure 16: Graph of Sub-hypercube

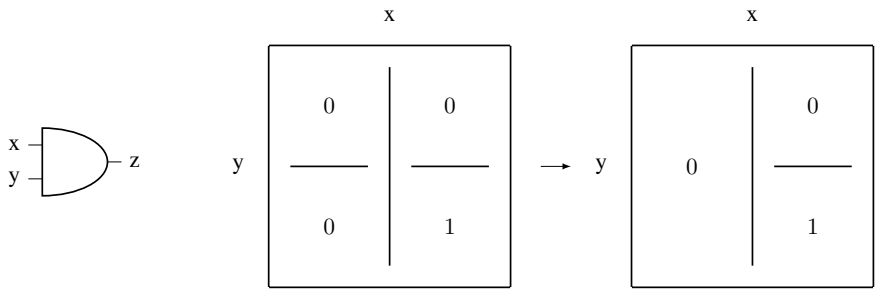


Figure 17: AND Gate

The OR gate (Figure 18) is 10 bits:

$$VHOOO, 0, 1, 1.$$

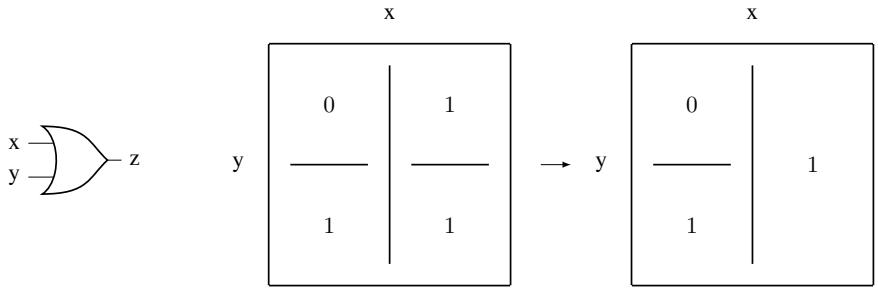


Figure 18: OR Gate

The XOR (Figure 19) gate is represented by 14 bits:

$$VHOOHOO, 0, 1, 1, 0.$$

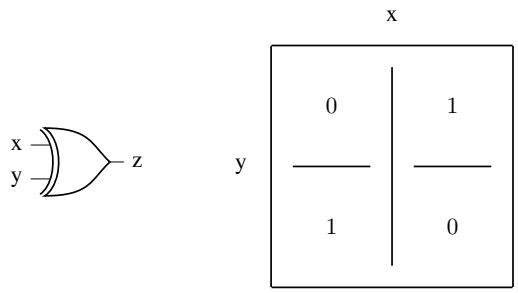


Figure 19: XOR Gate

The multiplexer (Figure 20) is represented by:

$$VOO, x, y.$$

The SR-Latch (Figure 21) is as follows:

$$VOO, \overline{q}, 0$$

and

$$VOO, q, 0.$$

The full adder is composed of the sum (Figure 22):

$$VHVOOVOOHVOOVOO, 0, 1, 1, 0, 1, 0, 0, 1$$

and carry (Figure 23):

$$VHVOOOHOVOO, 0, 1, 1, 0, 0, 1$$

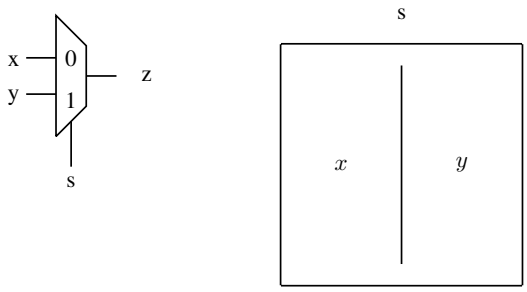


Figure 20: MUX Gate

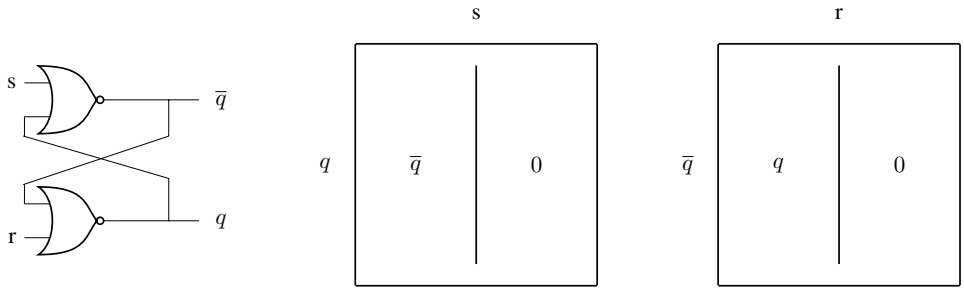


Figure 21: SR Gate

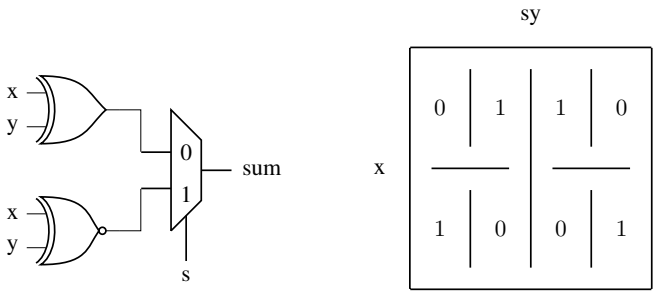


Figure 22: SUM Gate

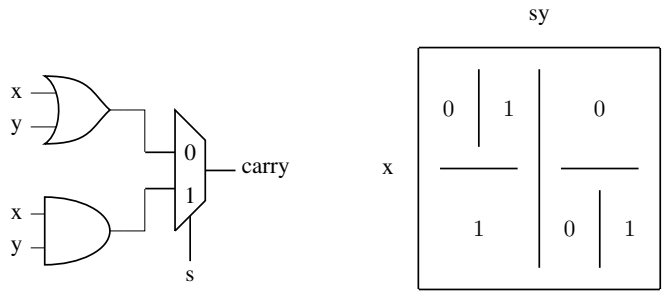


Figure 23: CARRY Gate

1.6 TERNARY- (TRINARY-) BASED LOGIC

The XOO notation allows three values for bisecting layouts: a ‘H’ or a ‘V’ for horizontal and vertical bisecting respectively and an ’O’ to illustrate a node.

Ternary is a base three numeral system using three digits: 0, 1, 2.

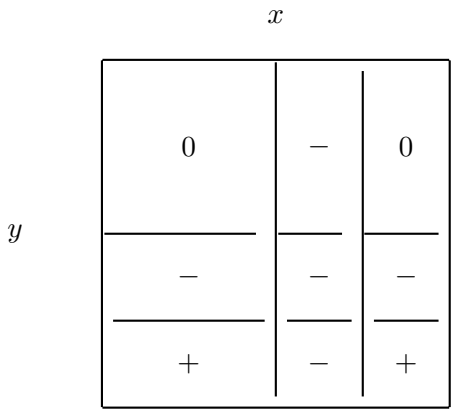
The balanced ternary notation uses the notation for values negative one, zero and positive one: -, 0, +. The balanced ternary notation will be used here.

The XOO notation implies a ternary numbering system. Ternary logical tables suggest that there may be a way to represent logic using the same XOO layout system. Unary and binary logical operators follow for ternary values:

x	NOT(x)
-	+
0	0
+	-

AND	-	0	+
-	-	-	-
0	-	0	0
+	-	0	+

The challenge is to derive a XOO layout that can represent the logical operators, sum-of-products and XOO encoding. The layout below is a general-purpose layout for ternary operators:



The difference between the ternary layout and previous binary layouts is that in the ternary layout the subdivisions are bisected between the zero and the nonzero values, and the nonzero value is then bisected between the negative and positive values. The bisection of the ternary layout does not alternate, either. The XOO encoding is:

$$VOHOHOOVHOHOOHOHOO, 0, -, +, 0, -, +, 0, -, +$$

Where V is +, H is - and O is 0:

$$+0-0-00+-0-00-0-000-+0-+0-+$$

In addition to the above stipulations for ternary layouts, a special notation is introduced because the outputs can take two possible values that contribute to a sum-of-products. This is where ternary layouts differentiate from the above alternating bisected layouts but it is covered as it provides a completeness to the overall subject area.

$$|x| = \begin{cases} + & \text{if } x = - \\ Don't\ Care & \text{if } x = 0 \\ + & \text{if } x = + \end{cases}$$

$$\lceil x \rceil = \begin{cases} - & \text{if } x = - \\ DC & \text{if } x = 0 \\ + & \text{if } x = + \end{cases}$$

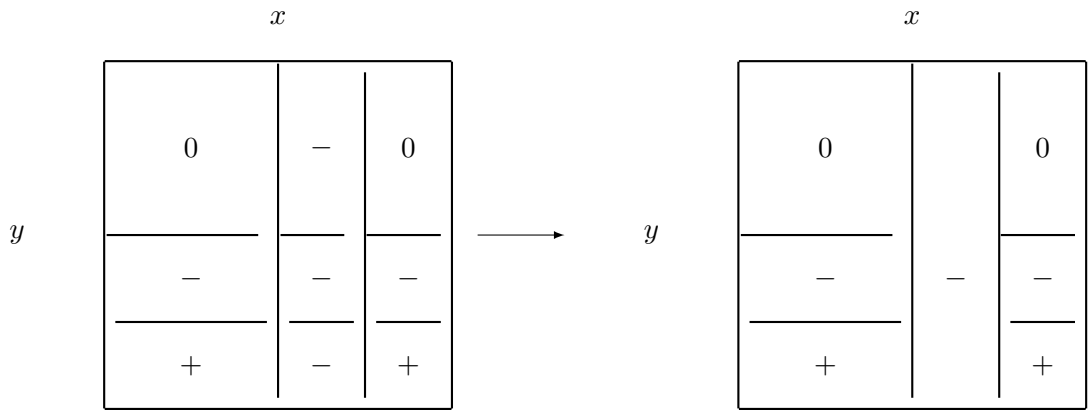
$$\lfloor x \rfloor = \begin{cases} - & \text{if } x = - \\ DC & \text{if } x \neq - \end{cases}$$

$$\lceil x \rceil = \begin{cases} + & \text{if } x = + \\ DC & \text{if } x \neq + \end{cases}$$

$$\overline{x} = \begin{cases} + & \text{if } x = 0 \\ DC & \text{if } x \neq 0 \end{cases}$$

$$\tilde{x} = \begin{cases} + & \text{if } x = - \\ DC & \text{if } x = 0 \\ - & \text{if } x = + \end{cases}$$

Taking the AND logical table above and simply transferring the values to the left layout below, subdivisions can be merged to show subdivisions spanning ‘y’. Then values are read off the simplified layout using the previous notation. The second column is ‘-1’ so it is ‘⌊x⌋’. The second row is ‘⌊y⌋’. Finally, the intersection of the third column and third row is ‘⌈x⌉⌈y⌉’.



The sum-of-products is:

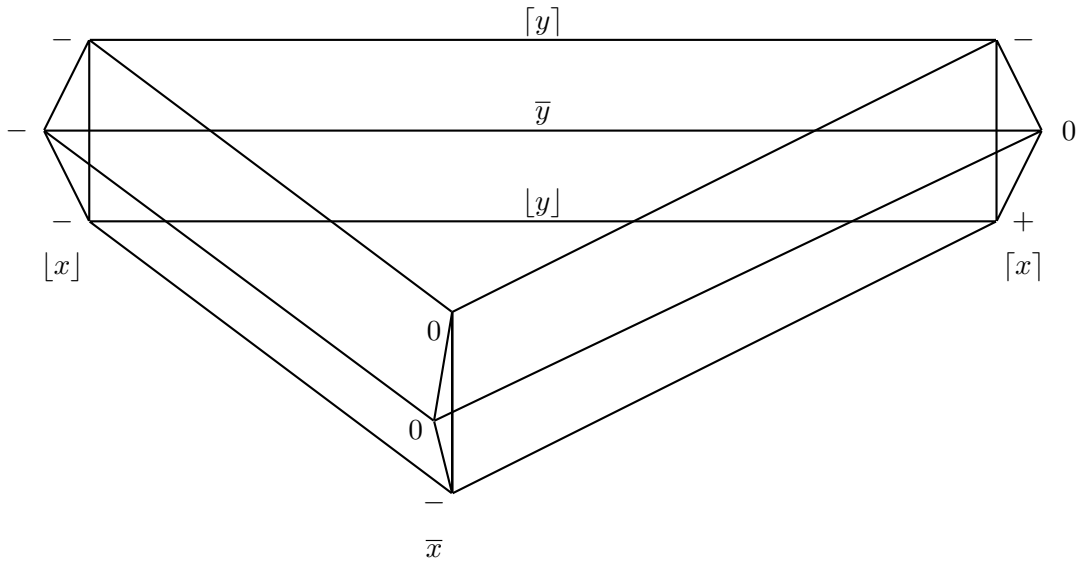
$$x \text{ AND } y = min(x, y) = \lfloor x \rfloor + \lfloor y \rfloor + \lceil x \rceil \lceil y \rceil$$

The ternary XOO encoding is:

$$VHOHOOVOHOHOO, 0, -, 0, -, 0, -, +$$

Or:

$$+ - 0 - 00 + 0 - 0 - 000 - 0 - 0 - +$$



1.7 GRAMMAR-BASED CODES

Suppose a XOO-encoding is given. A sub-encoding can be used to represent a portion of the original encoding, extracted and repeated. It can then be subsequently reversed to obtain the original encoding. A symbol table is applied to this process.

$x \rightarrow$ VOO
$\tilde{x} \rightarrow$ HOO

Only one symbol is needed with the tilde operator. Begin by substituting the symbol and extract each node 'O'.

VHVOOVOOHVOOVOO
VHxxHxx (substitute)
VHOOHOO,x,x,x,x (extract)
V$\tilde{x}\tilde{x}$,x,x,x,x (substitute)
VOO,$\tilde{x}$,$\tilde{x}$,x,x,x,x (extract)
x,$\tilde{x}$,$\tilde{x}$,x,x,x,x (end)

The commas can be removed from the ending result to obtain an efficient representation of the original string.

$x\tilde{x}\tilde{x}xxxx$

The opposite is possible. Begin with the first symbol and substitute its value.

x,$\tilde{x}$,$\tilde{x}$,x,x,x,x (add commas)
VOO,$\tilde{x}$,$\tilde{x}$,x,x,x,x (symbol)
V$\tilde{x}\tilde{x}$,x,x,x,x (insert)
VHOOHOO,x,x,x,x (symbol)
VHxxHxx (insert)
VHVOOVOOHVOOVOO (end)

1.8 CONTEXT-FREE GRAMMAR

The following is a context-free grammar for the XOO-encoding:

S $\rightarrow$ O
S $\rightarrow x\tilde{o}$
$x \rightarrow$ H|V
$\tilde{x} \rightarrow x\tilde{o}$
$o \rightarrow$ O
$\tilde{o} \rightarrow oo|x\tilde{o}|o\tilde{x}$

An optional additional rule is $\tilde{o} \rightarrow \tilde{x}\tilde{x}$.

It can be shown using the pumping lemma that a p-length XOO-encoding is non-regular. Given any value p a valid encoding xyz in the language $\mathcal{L}$ is one that satisfies these three properties:
$|y| >= 1$
$|xy| <= p$
$\forall n >= 0, xy^n z \in \mathcal{L}$
By pumping y, the encoding is no longer acceptable by the language $\mathcal{L}$.
Therefore, the XOO-encoding is a non-regular language.

Theorem 1.1. *The XOO-encoding is a non-regular language.*

Proof. The language $\mathcal{L}$ is the XOO-encoding and assume that it is regular.

A valid XOO string is $X^nO^{n+1} \in \mathcal{L}\ \forall n >= 0$

However, for $n = 0$ the string O cannot be composed of xyz.

For $n = 1$, the string XOO is composable into xyz.

This satisfies that $|y| >= 1$ and $|xy| <= p$.

Given a XOO-encoding it is composed of $x =$ X, $y =$ O and $z =$ O, i.e. $n = 1$.

Let $n = 2$. Then immediately the resulting generated encoding is $XOOO$ which is not in $\mathcal{L}$.

Therefore the XOO-encoding is non-regular. □

1.9 PARALLEL COMPUTING

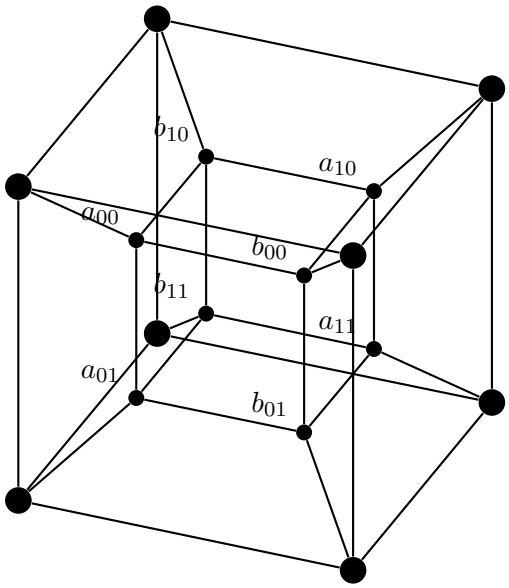
Parallel computing is a sophisticated area of computer science. It is a well established and mature technology.

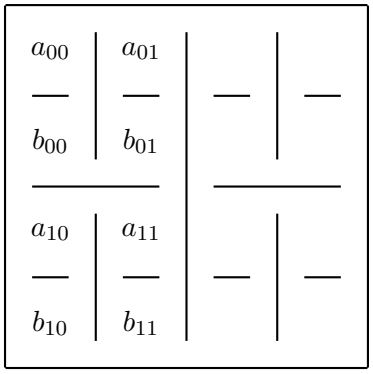
"Graphs having the topology of a hypercube [are] ideally suited to modeling distributed-memory parallel computers, mainly because of the fact that the diameter of a hypercube is the logarithm of the number of vertices." [59]

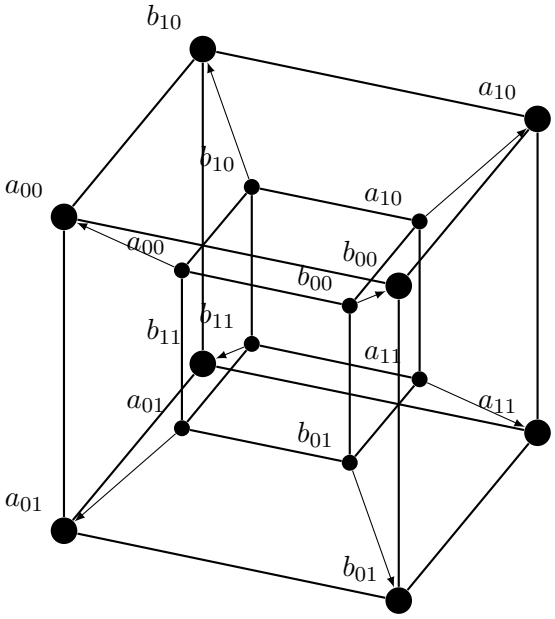
Hypercube algorithms exist to take advantage of multiple processing elements for many calculations at the same time. A simple way to illustrate this is through matrix multiplication.

$$AB = \begin{bmatrix} a_{00} & a_{01} \\ a_{10} & a_{11} \end{bmatrix} \begin{bmatrix} b_{00} & b_{01} \\ b_{10} & b_{11} \end{bmatrix} = \begin{bmatrix} a_{00}b_{00} + a_{01}b_{10} & a_{00}b_{01} + a_{01}b_{11} \\ a_{10}b_{00} + a_{11}b_{10} & a_{10}b_{01} + a_{11}b_{11} \end{bmatrix}$$

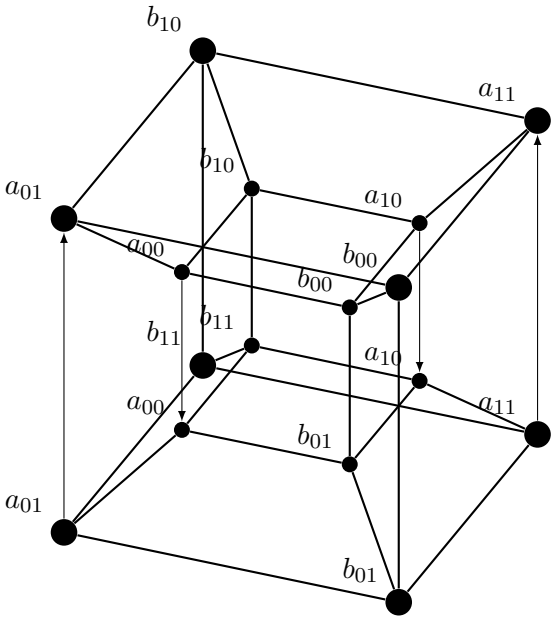
The process of matrix multiplication is done over multiple nodes - multiplication and addition are done through merging actions. The matrices are laid out in the XOO layout. Note that the hypercubes are simplified by one dimension in order to keep the illustration clearer, ie. the layout is 4D and the hypercube is 3D.



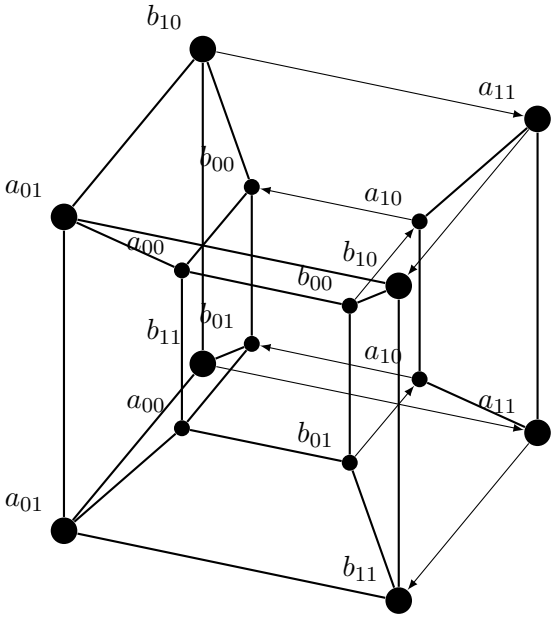




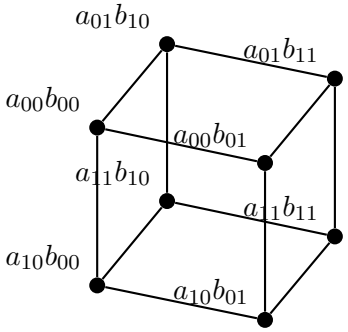
a_{00}	a_{01}	a_{01}	a_{00}
—	—	—	—
b_{00}	b_{01}	b_{01}	b_{00}
a_{10}	a_{11}	a_{11}	a_{10}
—	—	—	—
b_{10}	b_{11}	b_{11}	b_{10}



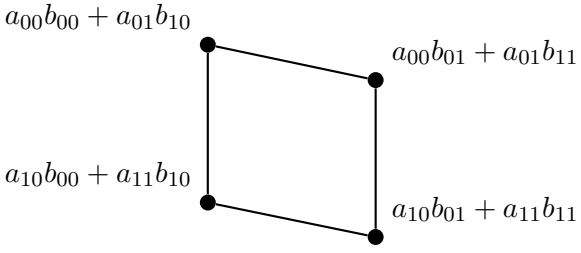
a_{00}	a_{00}	a_{01}	a_{01}
—	—	—	—
b_{00}	b_{01}	b_{01}	b_{00}
a_{10}	a_{10}	a_{11}	a_{11}
—	—	—	—
b_{10}	b_{11}	b_{11}	b_{10}



a_{00} / b_{00}	a_{00} / b_{01}	a_{01} / b_{11}	a_{01} / b_{10}
a_{10} / b_{00}	a_{10} / b_{01}	a_{11} / b_{11}	a_{11} / b_{10}



$a_{00}b_{00}$	$a_{00}b_{01}$	$a_{01}b_{11}$	$a_{01}b_{10}$
$a_{10}b_{00}$	$a_{10}b_{01}$	$a_{11}b_{11}$	$a_{11}b_{10}$



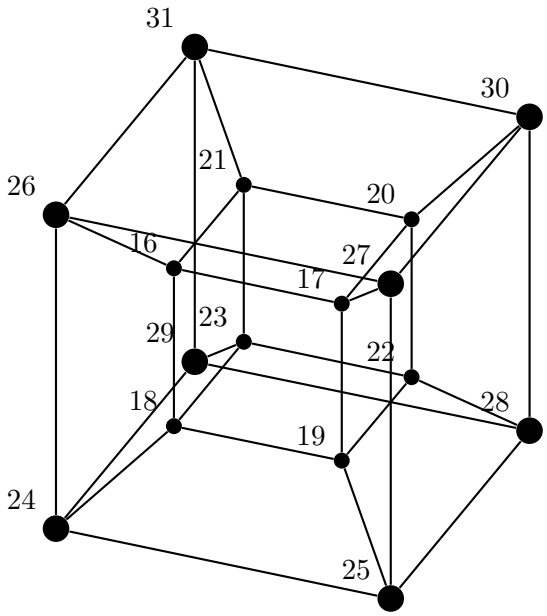
$a_{00}b_{00} + a_{01}b_{10}$	$a_{00}b_{01} + a_{01}b_{11}$
$a_{10}b_{00} + a_{11}b_{10}$	$a_{10}b_{01} + a_{11}b_{11}$

1.10 DISTRIBUTED SYSTEMS

There are two distributed systems that are of keen interest with respect to hypercube architecture and therefore the XOO layout schema. Content-addressable network (CAN) uses a layout scheme to

illustrate how nodes of a network are distributed. Kademlia is a distributed network that uniquely uses XOR as a distance metric using binary trees. [9] [10]

There are advantages in using the XOO layout scheme as a basis for a distributed system. It can be shown that CAN and Kademlia share properties using an implementation of a XOO schema in a distributed system. The XOR metric in Kademlia is used as a distance metric whereas when implemented in XOO it is seen as a dimensional measure.



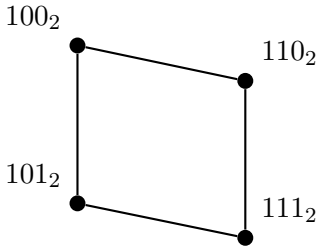
Nodes of a XOO layout in a distributed system represent clients/servers on a network. The XOO schema also has the advantage of determining which node stores/retrieves a hashed addressable content.

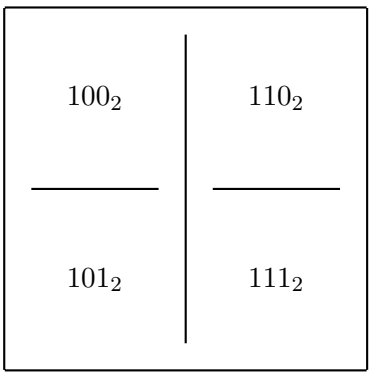
For example, given the string “xoo” that is SHA-1 hashed:

af633cd7b700e4832f05c2df6aaca651040acc6116

The binary value of the hash is:

$$101011110110...0001_2$$





Given a 0D network layout as the first case, there is only one node in the network that stores and retrieves all hashed content beginning with both 0_2 or 1_2 (0x0-0xF). Node 1_2 stores/retrieves the above hash.

Given a 1D dense layout, there are two nodes, 10_2 and 11_2. Node 10_2 captures all hashes beginning with 0_2 (or 0x0-0x7). Node 11_2 captures all hashes beginning with 1_2 (or 0x8-0xF) and stores/retrieves the above hash..

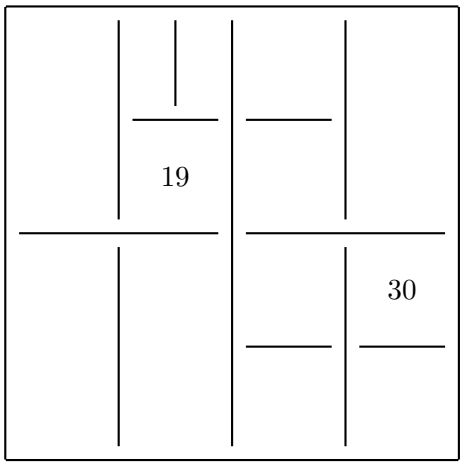
Given a 2D dense layout, there are four nodes: 100_2, 101_2, 110_2 and 111_2. 100_2 stores and retrieves hashes with values beginning with 00_2 (0x0-0x3), 101_2 stores/retrieves hashes with values beginning with 01_2 (0x4-0x7), and so on. Node 110_2 stores/retrieves the above hash.

The general rule is: given the node ID, the first bit of the node ID is ignored and the rest of the bits store and retrieve content addressable hashes matching the first set of bits of the hash. Given a hash, a network node determines whether it stores/retrieves the hashed content or passes the request along to a neighbourly node.

Now that it is known which node stores which hash, it is a matter of traversing the network from some node in the network to the destination node. This is done with a layout representation where each subdivision represents a node and each node knows only itself and its immediate neighbours.

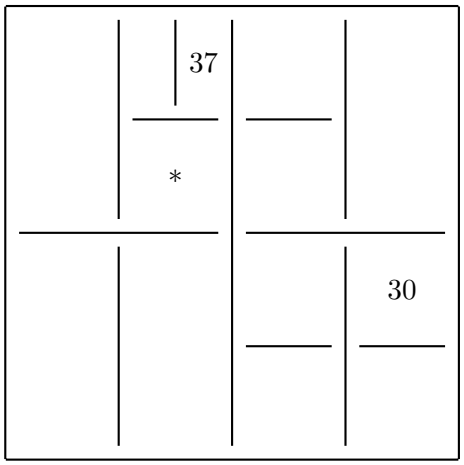
Node 19_{10} is queried for a hash beginning with 0xF8. The node calculates that it does not store those series of hashes, and calculates that none of its neighbours do, either.

Node 19_{10} then chooses some neighbour to further query.



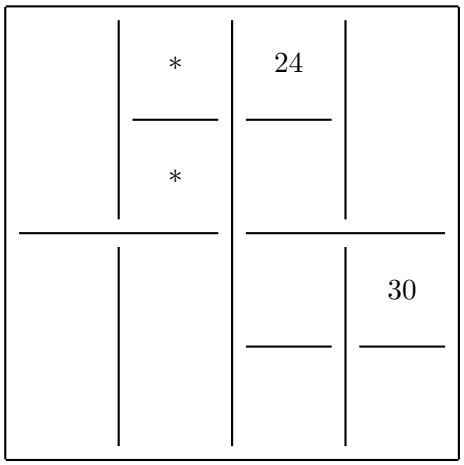
Node 37_{10} has been chosen and node 19_{10} has been flagged as the previous node. Node 37_{10} does not store the hash nor do its neighbours so one of its neighbours is chosen as the next node to query.

Node 19_{10} is not a candidate node since it was the previous node.

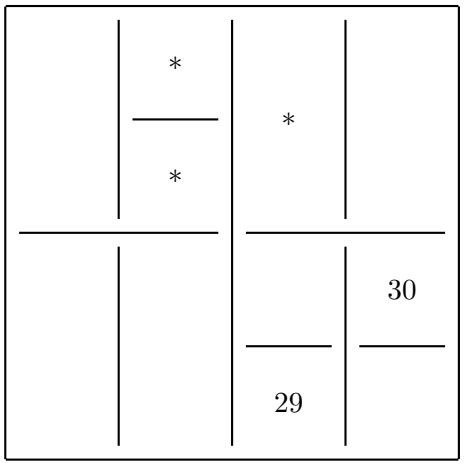


Node 24_{10} is the next node that has been chosen. It does not store the hash.

Since 37_{10} was the previous node, it is not flagged but instead removed from the mapping.* Node 24_{10} queries one of its neighbours in the mapping. Note that node 37_{10} is removed from the mapping which has been replaced by 18_{10} and 18_{10} is not a neighbour of 24_{10} in the original network layout.

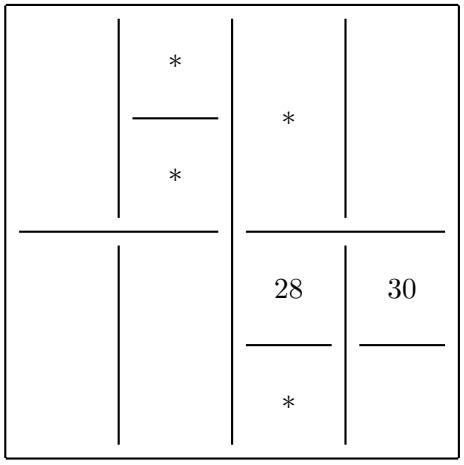


Node 29_{10} is chosen and it also does not store the hash nor do any of its neighbours. A neighbour is chosen randomly for the new query.



Node 28_{10} has been chosen. Node 29_{10} is flagged as a previous node. Node 30_{10} is identified as being a store for the hash.

Node 30_{10} is queried for the hash and the node returns its store.



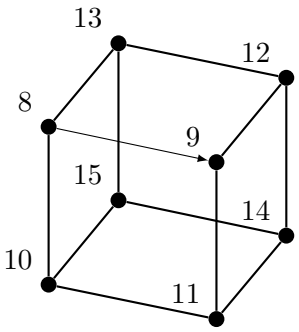
* A previous node is removed from the layout mapping when the new node chosen is not found transversally down the binary tree and is not its sibling.

1.11 SNAKE-IN-THE-BOX PROBLEM

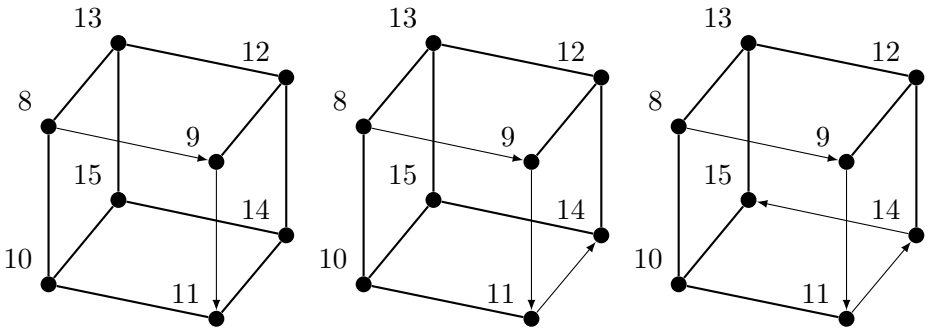
The snake-in-the-box problem is one in which to find the longest path in a hypercube where the path complies with certain attributes. The path traverses nodes where the beginning and ending nodes have only one neighbour on the path otherwise every node on the path has exactly two neighbours on the path.

The SIB problem is considered difficult as the hypercube increases in dimension. The number of possible choices that eliminate future possible paths becomes intractable while trying to maximize the path length through the hypercube. [83]

It's now proposed that an alternating-oriented bisected layout can be used to solve this problem. First, the problem is illustrated using a three-dimensional cube.

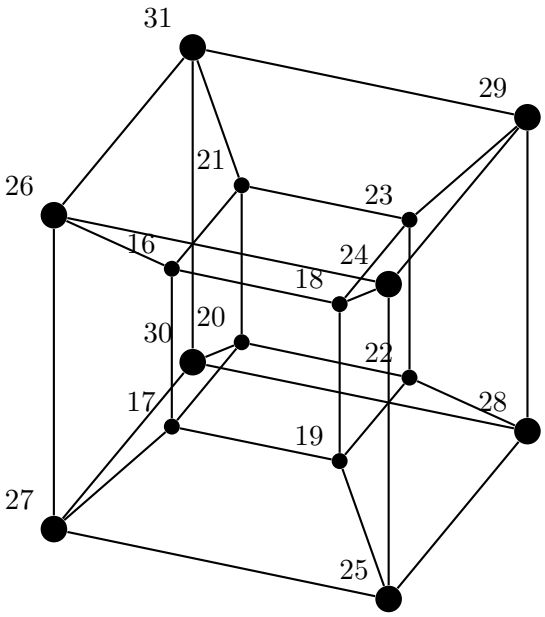


Starting a node 8, choose any node to traverse to, so node 9 is chosen. Eliminate (grey out) the other neighbour nodes of 8 since they cannot be chosen by any future path. Continue the process from node 9 to node 11.

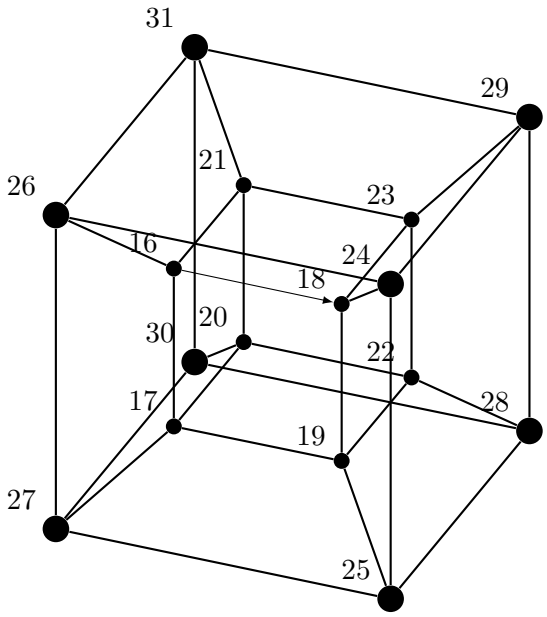


The maximal path length of a three-dimensional cube is 4 edges.

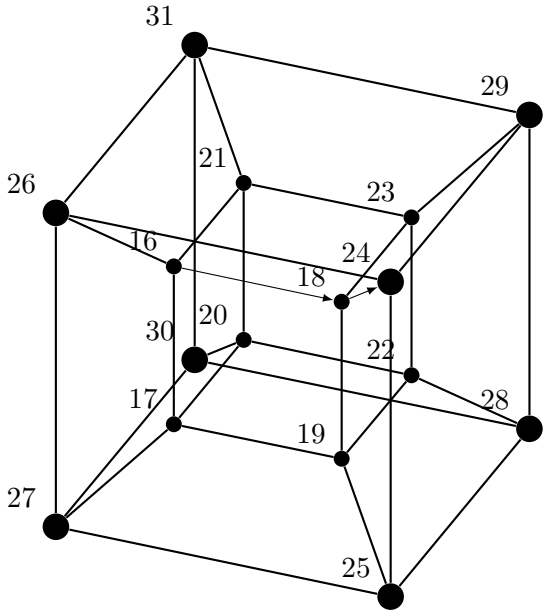
A four-dimensional hypercube maximal path will be illustrated with the benefit of an alternating-oriented bisected layout. In addition to illustrating the path on the hypercube, the layout will show the path from node-to-node and eliminated nodes are represented with an ‘x’.



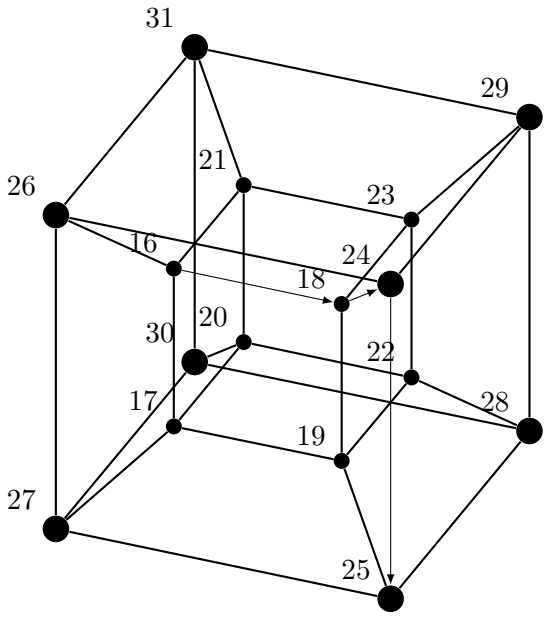
16	18	24	26
17	19	25	27
20	22	28	30
21	23	29	31

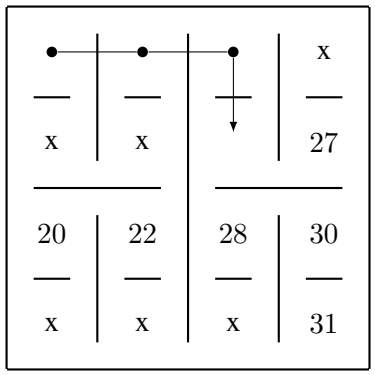


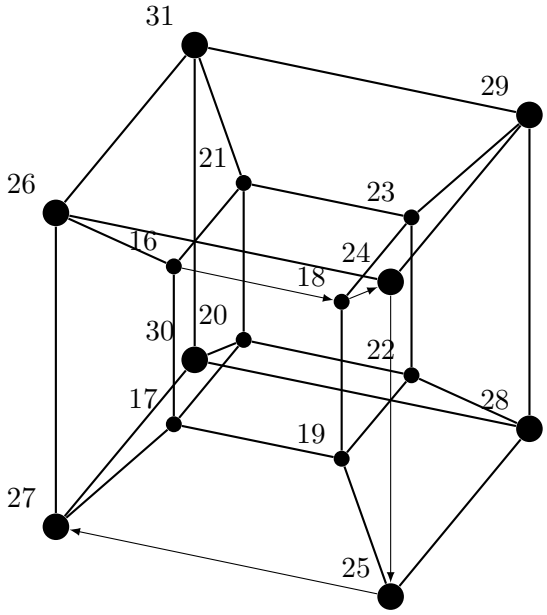
● →		24	x
x	19	25	27
20	22	28	30
x	23	29	31

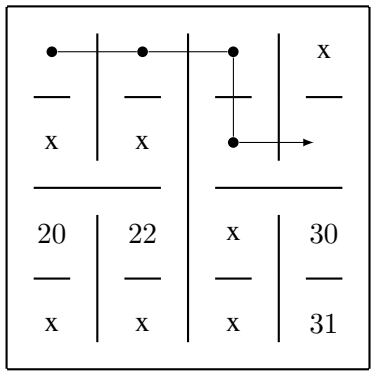


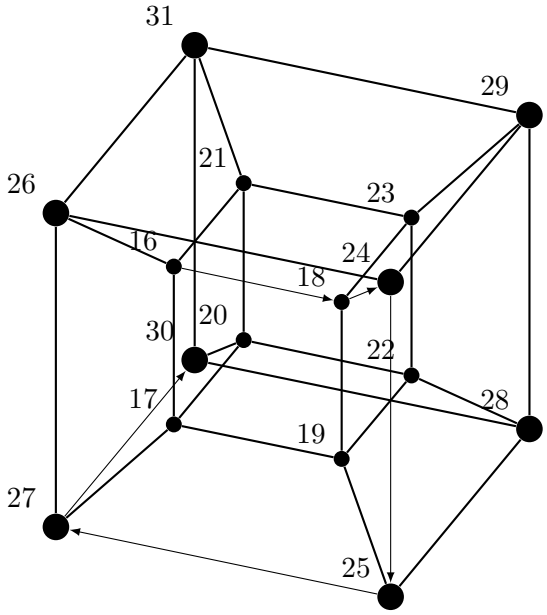
● → ● →			x
x	x	25	27
20	22	28	30
x	x	29	31

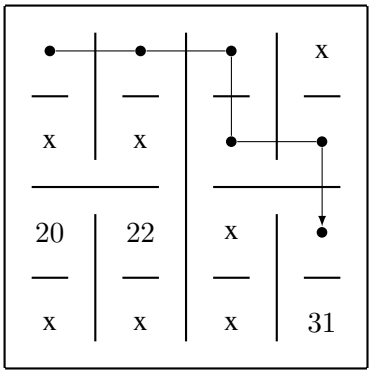

31
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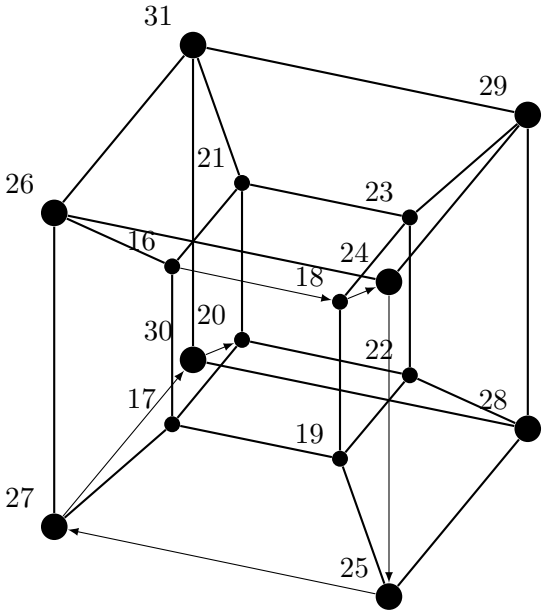

x
x
x
27
20
22
28
30
x
x
x
31

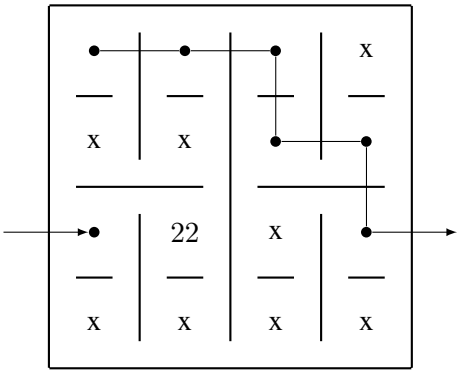

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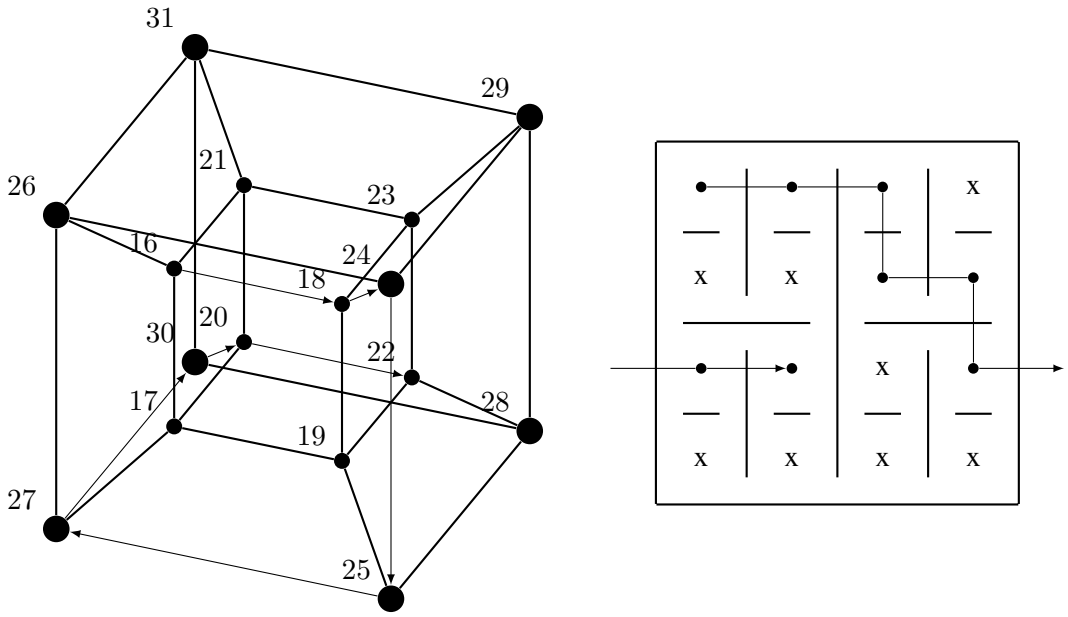

x
x
x
20
22
x
30
x
x
x
31


31
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x
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22
x
x
x
x
31


31
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16
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18
20
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28
19
27
25


x
x
x
22
x
x
x
x



The 4-dimensional hypercube has a maximal path of 7 edges. The value of the alternating-oriented bisected layout illustrates a useful data structure for problem-solving.

In addition, the layout is labelled "ABCD". The above particular maximal path can be algebraically represented as:

$$\overline{AD} + A\overline{BC} + A\overline{B}D + BC\overline{D}$$

With the use of the alternating-oriented bisected layout the problem space collapses. The following theorems of maximal paths over hypercubes were deduced.

Theorem 1.2. *The maximal path for hypercubes is:*

Even dimensions > 2:

$$e = (2^{\frac{d-2}{2}} - 1)(2^{\frac{d}{2}} + 3)$$

Odd dimensions > 3:

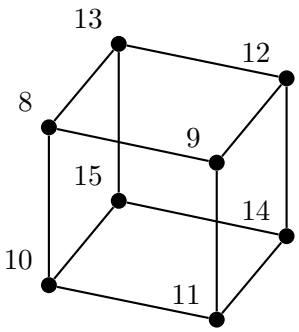
$$e = (2^{\frac{d-1}{2}} - 1)(2^{\frac{d-1}{2}} + 2)$$

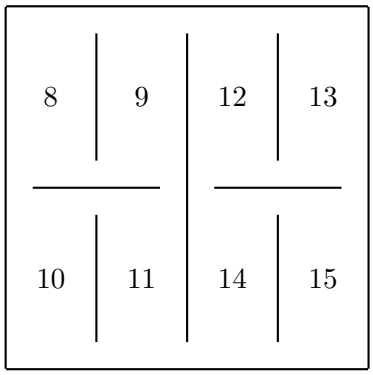
Theorem 1.3. *The maximal path on hypercube dimension d>2 is:*

$$e = (2^{\frac{d-(d-1) \bmod 2-1}{2}} - 1)(2^{\frac{d-d \bmod 2}{2}} + (d-1) \bmod 2 + 2)$$

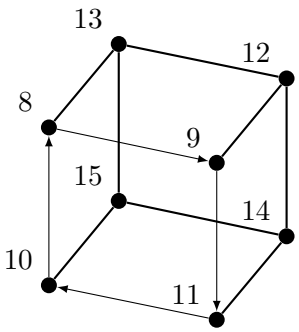
1.12 PERMUTATIONS

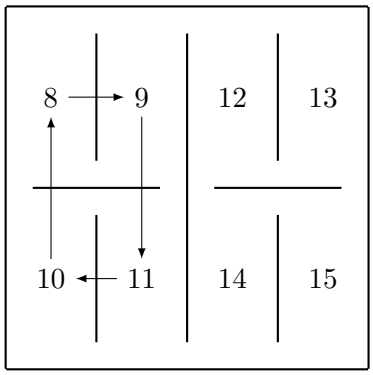
The three dimensional cube and the XOO layout can be theorized as a simplified Rubik's cube where each face of the cube can be rotated by permuting the labels of a face.

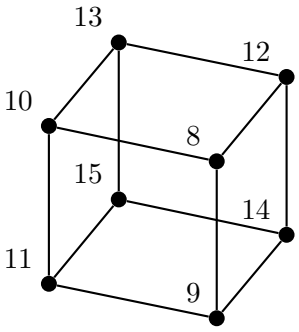


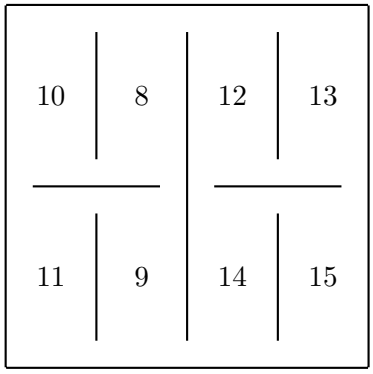


The above layout is the starting orientation. The following results in the 8-9-11-10 (a four-cycle) face being "rotated" relative to the starting orientation. There are six possible explicit single four-cycle rotations.

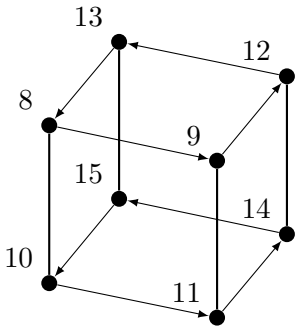


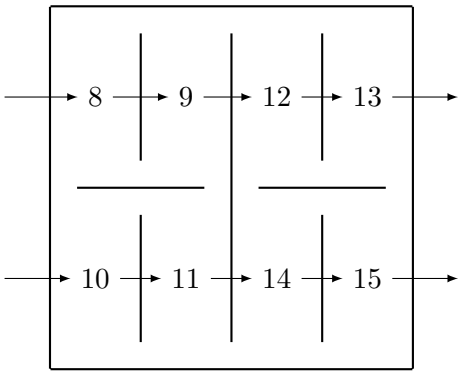


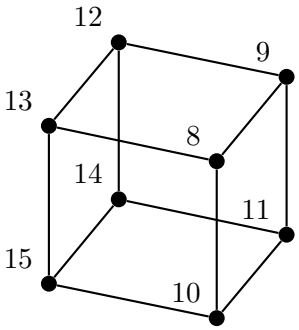


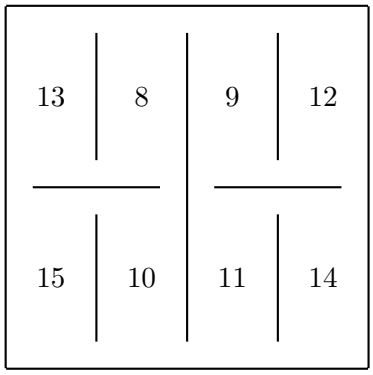


There are three possible implicit double four-cycle rotations as if the middle plane of a Rubik's cube is implied to rotate in the counter-direction.





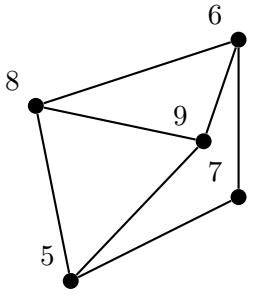




1.13 GRAPH OPERATIONS

Graph theory, the mathematics of vertex-edge sets, can be used to study operations on hypercube graphs. Matrix operations on matrices that represent graphs can add further room to study hypercubes and the XOO layout.

Since the XOO layout is used as a basis for hypercube graphs, the vertices are labelled according to the previous node labelling. Any operations on a graph that affect vertices also affect the labelling.

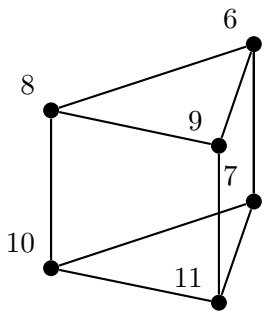


$$
\begin{array}{c} \\ 8 \\ 9 \\ 5 \\ 6 \\ 7 \end{array}
\begin{array}{c}
\begin{array}{ccccc} 8 & 9 & 5 & 6 & 7 \end{array} \\
\begin{bmatrix}
0 & 1 & 1 & 1 & 0 \\
1 & 0 & 1 & 1 & 0 \\
1 & 1 & 0 & 0 & 1 \\
1 & 1 & 0 & 0 & 1 \\
0 & 0 & 1 & 1 & 0
\end{bmatrix}
\end{array}
$$

$$
R_0GC_0 = \begin{bmatrix}
1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
\begin{bmatrix}
0 & 1 & 1 & 1 & 0 \\
1 & 0 & 1 & 1 & 0 \\
1 & 1 & 0 & 0 & 1 \\
1 & 1 & 0 & 0 & 1 \\
0 & 0 & 1 & 1 & 0
\end{bmatrix}
\begin{bmatrix}
1 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
R_1GC_1 = \begin{bmatrix}
0 & 0 & 0 & 0 & 0 \\
1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 1
\end{bmatrix}
\begin{bmatrix}
0 & 1 & 1 & 1 & 0 \\
1 & 0 & 1 & 1 & 0 \\
1 & 1 & 0 & 0 & 1 \\
1 & 1 & 0 & 0 & 1 \\
0 & 0 & 1 & 1 & 0
\end{bmatrix}
\begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 1 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 0 \\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\begin{array}{c} \\ 8 \\ 9 \\ 10 \\ 11 \\ 6 \\ 7 \end{array}
\begin{array}{c}
\begin{array}{cccccc} 8 & 9 & 10 & 11 & 6 & 7 \end{array} \\
\begin{bmatrix}
0 & 1 & 1 & 0 & 1 & 0 \\
1 & 0 & 0 & 1 & 1 & 0 \\
1 & 0 & 0 & 1 & 0 & 1 \\
0 & 1 & 1 & 0 & 0 & 1 \\
1 & 1 & 0 & 0 & 0 & 1 \\
0 & 0 & 1 & 1 & 1 & 0
\end{bmatrix}
\end{array}
$$



1.14 PROGRAMMING

An area of research still to be done is to develop a programming language based on hypercubes. As can be seen above, there is much interesting research and development to be done in different fields of

computer science and applications development with respect to the XOO schema. This paper provides the basis for new areas of research and development.

A hypercube programming language and a graph API can be developed as operations on the graph abstract data type [81] [82]:

adjacent(G, x, y): tests whether there is an edge from the vertices x to y;

neighbors(G, x): lists all vertices y such that there is an edge from the vertices x to y;

add_vertex(G, x): adds the vertex x, if it is not there;

remove_vertex(G, x): removes the vertex x, if it is there;

add_edge(G, x, y): adds the edge from the vertices x to y, if it is not there;

remove_edge(G, x, y): removes the edge from the vertices x to y, if it is there;

get_vertex_value(G, x): returns the value associated with the vertex x;

set_vertex_value(G, x, v): sets the value associated with the vertex x to v.

get_edge_value(G, x, y): returns the value associated with the edge (x, y);

set_edge_value(G, x, y, v): sets the value associated with the edge (x, y) to v.

In addition, a distributed network can provide an API to store/retrieve content:

ping: used to verify that a node is alive;

store: stores a (key, value) pair in one node;

find_node: the recipient of the request will return the k nodes closest to the requested key;

find_value: same as find_node but if the recipient of the request has the requested key in its store it will return the corresponding value;

A web application is being built using "a method for users to create and edit web page layouts" where every division (node) of a layout represents a network resource or process. The properties of the application demonstrate that the XOO layout schema is a viable technology on which to build efficient hypercube web applications and systems. [84]

A line of application development is a XOO layout API. Beginning with a text-based terminal, it would be trivial to display fixed-width text layouts and implement a simple text-based windowing system. Building on that, windows in a desktop environment can be laid out according to the XOO layout schema. Both textual and window environments would implement aspects of the XOO schema. This way, every division of a layout (ie. a node of a hypercube) can represent a network resource (distributed hypercube) or a process (parallel computing process) as mentioned in the above web application. The way data is presented, networked, processed and laid out would all be aspects of the XOO layout schema and, by extension, hypercube architecture.

Finally, the mathematics of hypercubes are well established but with the realization of the interchangeable data structures shown above there are real-world applications of mathematical theory throughout the XOO data structure. Putting the XOO data structure on a firm foundation will be done via a mathematical and computational survey. This paper illustrates the wide areas of symmetry to be pursued in research and development of the XOO schema and hypercubes.

1.15 TURING MACHINE

The following program verifies that a XOO-encoding is valid.

$M = (\Sigma, \Gamma, Q, \delta, q, q_a, q_r)$

$\Sigma = (H, V, O)$
$\Gamma = (\square, X, H, V, O)$
$Q = (q_{start}, q_{accept}, q_{reject}, q_x, q'_x, q''_x, q_0, q'_0, q''_0, q_{return}, q'_{return}, q_{merge}, q'_{merge}, q''_{merge},$
$q_{shift}, q_{shift_X}, q_{shift_O}, q'_{shift_O}, q_{shift_H}, q'_{shift_H}, q_{shift_V}, q'_{shift_V},$
$q_{print_O}, q_{print_O}, q_{print_H}, q'_{print_H}, q_{print_V}, q'_{print_V})$

$$
\delta = \begin{cases}
q_{start}\square \to q_{reject} & q'_{merge}O \to q''_{merge}\square R \\
q_{start}O \to q_0 OL & q''_{merge}O \to q_{shift}\square R \\
q_{start}H \to q_x HR & q_{print_O}O \to q_{print_O}\square L \\
q_{start}V \to q_x VR & q_{print_O}H \to q'_{print_O}OR \\
q_x\square \to q_{reject} & q_{print_O}V \to q'_{print_O}OR \\
q_xO \to q'_xOR & q'_{print_O}\square \to q_{shift_X}OR \\
q_xH \to q_xHR & q_{print_H}O \to q_{print_H}\square L \\
q_xV \to q_xVR & q_{print_H}H \to q'_{print_H}OR \\
q'_x\square \to q_{reject} & q_{print_H}V \to q'_{print_H}OR \\
q'_xO \to q''_xOR & q'_{print_H}\square \to q_{shift_X}HR \\
q'_xH \to q_xHR & q_{print_V}O \to q_{print_V}\square L \\
q'_xV \to q_xVR & q_{print_V}H \to q'_{print_V}OR \\
q''_x\square \to q'_{return}\square L & q_{print_V}V \to q'_{print_V}OR \\
q''_xO \to q_{print_O}XL & q'_{print_V}\square \to q_{shift_X}VR \\
q''_xH \to q_{print_H}XL & q_{shift}\square \to q_{return}\square L \\
q''_xV \to q_{print_V}XL & q_{shift}O \to q_{shift_O}XL \\
q_0\square \to q'_0\square R & q_{shift}H \to q_{shift_H}XL \\
q_0O \to q_{merge}OL & q_{shift}V \to q_{shift_V}XL \\
q_0H \to q_xHR & q_{shift_X}\square \to q_{shift_X}\square R \\
q_0V \to q_xVR & q_{shift_X}X \to q_{shift}\square R \\
q'_0O \to q''_0OR & q_{shift_O}\square \to q'_{shift_O}\square L \\
q''_0\square \to q_{accept} & q_{shift_O}O \to q'_{shift_O}OR \\
q''_0O \to q_{reject} & q_{shift_O}H \to q'_{shift_O}HR \\
q''_0H \to q_{reject} & q_{shift_O}V \to q'_{shift_O}VR \\
q''_0V \to q_{reject} & q'_{shift_O}\square \to q_{shift_X}OR \\
q_{return}\square \to q_{return}\square L & q_{shift_H}\square \to q'_{shift_H}\square L \\
q_{return}O \to q_{start}ON & q_{shift_H}O \to q'_{shift_H}OR \\
q_{return}H \to q_xHR & q_{shift_H}H \to q'_{shift_H}HR \\
q_{return}V \to q_xVR & q_{shift_H}V \to q'_{shift_H}VR \\
q'_{return}O \to q'_{return}\square R & q'_{shift_H}\square \to q_{shift_X}HR \\
q'_{return}H \to q'_0ON & q_{shift_V}\square \to q'_{shift_V}\square L \\
q'_{return}V \to q'_0ON & q_{shift_V}O \to q'_{shift_V}OR \\
q_{merge}\square \to q_{reject} & q_{shift_V}H \to q'_{shift_V}HR \\
q_{merge}O \to q_{merge}OL & q_{shift_V}V \to q'_{shift_V}VR \\
q_{merge}H \to q'_{merge}OR & q'_{shift_V}\square \to q_{shift_X}VR \\
q_{merge}V \to q'_{merge}OR &
\end{cases}
$$

2 Definition of $Exclusive\text{-}One$

Definition 2.1. *Exclusive-One - Exclusive-one is true when only one symbol is true and undefined otherwise.*

The exclusive-one is a sum-of-products where XOR operates on each pairwise symbols on each successive term with the first pair of the first term and ending with the last positive single symbol or pair of symbols of the last term.

Generally,

$$\Gamma(n) = \begin{cases} \Gamma_0 & \text{if } n \pmod 2 = 0 \text{ and } n = 0 \\ \prod_{p=0}^{n} \overline{\Gamma}_p \sum_{m=0}^{\frac{n-1}{2}} (\Gamma_{2m} \oplus \Gamma_{2m+1}) \Gamma_{2m} \Gamma_{2m+1} & \text{if } n \pmod 2 = 1 \\ \Gamma_{n-1} \overline{\Gamma}_n + \prod_{p=0}^{n-1} \overline{\Gamma}_p \Gamma_n & \text{if } n \pmod 2 = 0 \end{cases}.$$

Logically, one representation is

$$\Gamma(3) = (\Gamma_0 \oplus \Gamma_1)\overline{\Gamma}_2\overline{\Gamma}_3 + \overline{\Gamma}_0\overline{\Gamma}_1(\Gamma_2 \oplus \Gamma_3).$$

Alternatively,

$$\Gamma(n) = \prod_{p=0}^{n} \overline{\Gamma}_p \sum_{m=0}^{n} \Gamma_m \Gamma_m.$$

An alternate way of representing an exclusive-one is as a positive contribution progressing per each of n-terms. For example,

$$\Gamma(3) = \Gamma_0\overline{\Gamma}_1\overline{\Gamma}_2\overline{\Gamma}_3 + \overline{\Gamma}_0\Gamma_1\overline{\Gamma}_2\overline{\Gamma}_3 + \overline{\Gamma}_0\overline{\Gamma}_1\Gamma_2\overline{\Gamma}_3 + \overline{\Gamma}_0\overline{\Gamma}_1\overline{\Gamma}_2\Gamma_3.$$

Either of the above exclusive-one operations suffices to map to an alternating-oriented layout where only an exclusive-one is true and all others are zero. Non-exclusive-one operations are undefined.

$$\delta : (Q \backslash F)\Gamma \mapsto Q\Gamma\{L, R\}.$$

By applying the partial function over the XOO-layout, each exclusive-one Γ_n is mapped to a unique subdivision represented by an expanded binary string equal to its mapping. The partial function maps each Γ_n to a unique subdivision and therefore its result. Suppose that Γ_2 in Γ is to be mapped. Alternate along the orientation while reading the symbols $left - to - right$ and selecting the $left$ or top subdivision whenever the symbol is $zero$, and $right$ and $bottom$ when the symbol is one until all symbols are traversed.

In figure 24, $\Gamma_0 \mapsto \Gamma_{1000}$, $\Gamma_1 \mapsto \Gamma_{0100}$, $\Gamma_2 \mapsto \Gamma_{0010}$ and $\Gamma_3 \mapsto \Gamma_{0001}$.

The XOO-layout can be configured a number of ways. Figure 24 shows a mapping where Γ maps to subdivisions along the edge which is reflected in the binary string subscript. Figure 25 shows a mapping such that each subdivision alternates in a zipper fashion towards the $left - top$ corner. Each mapping configuration will be described by unique expanded binary strings.

In figure 25, $\Gamma_0 \mapsto \Gamma_{0^01}$, $\Gamma_1 \mapsto \Gamma_{01}$, $\Gamma_2 \mapsto \Gamma_{001}$ and $\Gamma_3 \mapsto \Gamma_{0001}$.

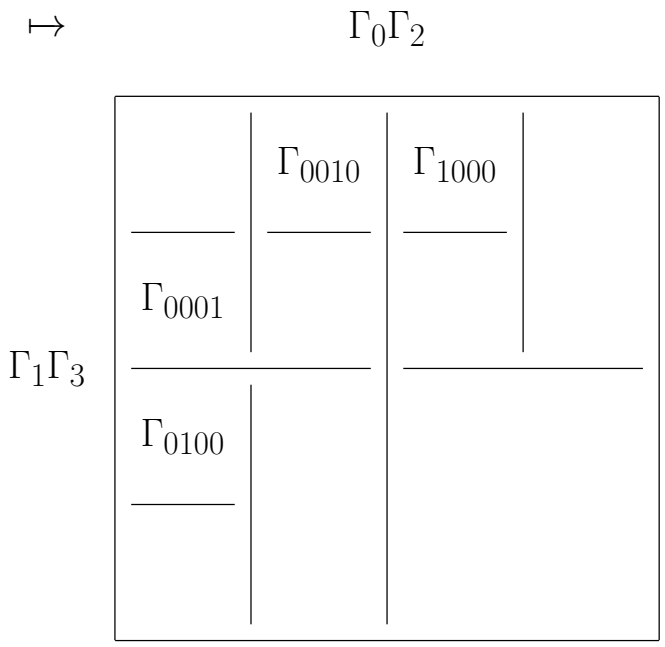


Figure 24: $\Gamma_n \mapsto \Gamma_{0^n 10^{|\Gamma|-n-1}}$ where the nth bit in Γ is set.

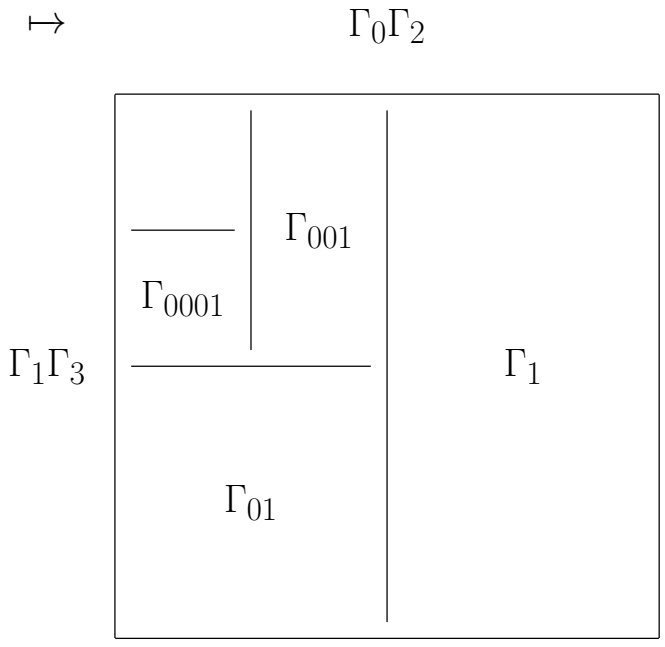


Figure 25: $\Gamma_n \mapsto \Gamma_{0^n 1}$

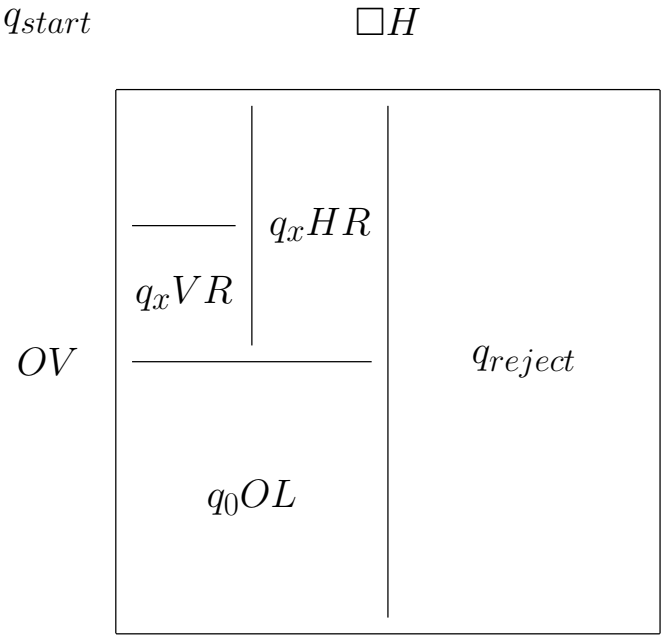


Figure 26: q_{start}

Figure 26 introduces the partial function mapping of the Turing machine. Each XOO-layout is a state space of exclusive-one symbols mapping to subsequent $Q\Gamma\{L, R\}$. The exclusive-one symbol is read by the tape head for the current state as labelled above the $top - left$ of the XOO-layout. In the figure, $\Gamma_0 \equiv \square$, $\Gamma_1 \equiv O$ and so on.

Applying this logic results in Appendix A which is a δ-collection of XOO-encodings of the Turing machine in section 1.5.

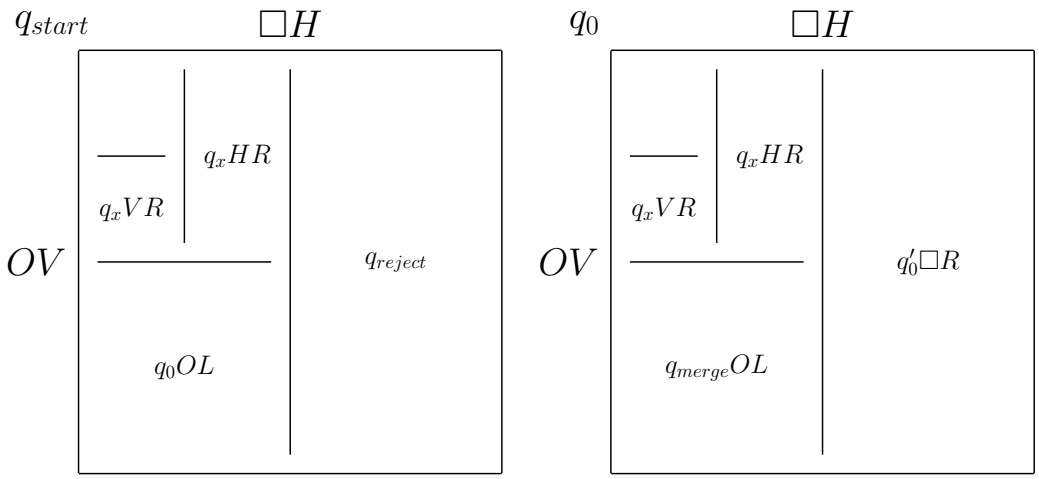
3 CONCLUSION

Alternating-oriented bisected layouts have been demonstrated to be an efficient means of representing logic with a connection to graphs and layouts of hypercubes. It has been shown that it is an efficient method compared to Karnaugh maps.

The XOO-encoding grammar has also been proven to be non-regular (Section 1.8). The Turing machine validates that a given encoding is a valid XOO-encoding (Section 1.15). The Turing machine is represented as XOO-encodings (Appendix A). Therefore, the XOO-encoding is Turing complete.

A APPENDIX

The following δ collection of XOO-encodings is equivalent to the Turing machine in section 1.5.


q_{start}
$\square H$
q_xHR
q_xVR
OV
q_{reject}
q_0OL
q_0
$\square H$
q_xHR
q_xVR
OV
$q_0'\square R$
$q_{merge}OL$

A.1 TERNARY LOGIC TABLE

-	0	+	x	UNARY OPERATOR
-	-	-	$\tilde{\bar{x}} + \lfloor \tilde{x} \rfloor$	
-	-	0	$\tilde{\bar{x}} + \lfloor x \rfloor$	
-	-	+	$\tilde{\bar{x}} + \lceil x \rceil$	Is True
-	0	-	$\lfloor \tilde{x} \rfloor$	
-	0	0	$\lfloor x \rfloor$	* Clamp Down
-	0	+	$\lceil x \rceil$	* Buffer
-	+	-	$\bar{x} + \lfloor \tilde{x} \rfloor$	Is Unknown
-	+	0	$\bar{x} + \lfloor x \rfloor$	
-	+	+	$\bar{x} + \lceil x \rceil$	
0	-	-	$\tilde{\bar{x}} + \lceil \tilde{x} \rceil$	
0	-	0	$\tilde{\bar{x}}$	
0	-	+	$\tilde{\bar{x}} + \lceil x \rceil$	
0	0	-	$\lceil \tilde{x} \rceil$	
0	0	0		Don't Care
0	0	+	$\lceil x \rceil$	* Clamp Up
0	+	-	$\bar{x} + \lceil \tilde{x} \rceil$	Increment
0	+	0	$\bar{x}$	*
0	+	+	$\bar{x} + \lceil x \rceil$	
+	-	-	$\tilde{\bar{x}} + \lfloor \tilde{x} \rfloor$	Is False
+	-	0	$\tilde{\bar{x}} + \lfloor \tilde{x} \rfloor$	Decrement
+	-	+	$\tilde{\bar{x}} + \lfloor x \rfloor$	
+	0	-	$\lfloor \tilde{x} \rfloor$	Not
+	0	0	$\lfloor \tilde{x} \rfloor$	
+	0	+	$\lfloor x \rfloor$	*
+	+	-	$\bar{x} + \lfloor \tilde{x} \rfloor$	
+	+	0	$\bar{x} + \lfloor \tilde{x} \rfloor$	
+	+	+	$\bar{x} + \lfloor x \rfloor$	

* Defined ternary operators [33]

A.2 KEYWORDS

- Actor Model
- Algebra
  - Terwilliger Algebra
- Algorithm
- Architecture
- Byzantine Generals Problem
- Computing
  - Distributed
  - Parallel
- Cryptography
- Cubical Calculus
  - Cube Notation
- Elections
- Finite State Machine
- Fuzzy (Logic)
  - Fuzzy Cube
- Genetics
  - Hypercube polynucleotide
- Grammar-based Codes
- Graph
  - Abstract Data Type
- Group
  - Hyperoctahedral Group
- Index (Search) Trees
  - K-D-B-tree (k-dimensional b-tree)
  - Bkd-tree
- Latin Hypercube Sampling
- Layout

  - Circuit
  - Floorplan
  - VLSI (Very Large Scale Integration)
- Linguistics
  - Geometrical Morphology
- Measure Polytope Hypercube Sequence
- MIMD (Multiple Instruction Multiple Data)
- Nanocomputer
  - M-Hypercube
- Network
  - Decentralized
  - Distributed
  - Interconnection
- OLAP (Online Analytical Processing)
  - Hypertable
- P2P (Peer-to-peer)
  - CAN (Content-Addressable Network)
  - Kademlia
- Plan 9
- Routing
- Semantic Web
- Snake-in-the-Box
- Space-Partitioning Trees
  - 2D Tree
  - K-d Tree
  - K-D-B Tree
- Switching Theory
- Ternary
- Tesseract
- Topology
  - Torus
- Voxel

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